

Assignment 2

Object Detection with YOLO11n on Skin Disease Images

Image and Video Analysis with Deep Learning | SS 2026

Narseyit Shyngys Malikov Fakhri Golovatenko Vladislav

May 2026

Dataset

Source: Roboflow Universe — skin-diseases-iyitj/skin-k1hbx v4 (License: CC BY 4.0)

Domain: Clinical and dermoscopic skin lesion photography — visually distinct from the 80 MS COCO classes (people, vehicles, household objects) used to pre-train YOLO11n, making it a meaningful domain-adaptation challenge.

Classes (2): Eczema melanoma-disease

Split	Train	Validation	Test
Images	145	41	21

Table 1: Dataset split sizes (207 images total).

Model Selection

Model: YOLO11n (Ultralytics, nano variant) **Pre-trained on:** MS COCO 2017 (80 classes)

Justification: YOLO11n (≈ 2.6 M parameters, 5.4 MB weight file) is the smallest variant of the YOLO11 family. It trains feasibly on consumer hardware using Apple MPS acceleration within the time budget. Compared to YOLOv8n, YOLO11 features an improved C2PSA attention block and a redesigned neck, yielding a small mAP gain at the same inference speed. The nano size ensures batch=16 at 640 px fits comfortably in memory without out-of-memory risk.

Baseline Results (Pre-trained, No Fine-Tuning)

The pre-trained YOLO11n was applied to all 21 test images at a confidence threshold of 0.25, without any adaptation to the skin-disease domain.

Because YOLO11n’s 80 COCO classes contain no skin-disease categories, the model produces **no valid detections**: it occasionally fires spurious predictions on structurally similar COCO objects (e.g., textures resembling *sports ball* or *frisbee*) but at very low confidence. The formal mAP50 for **melanoma-disease** is exactly 0.000.

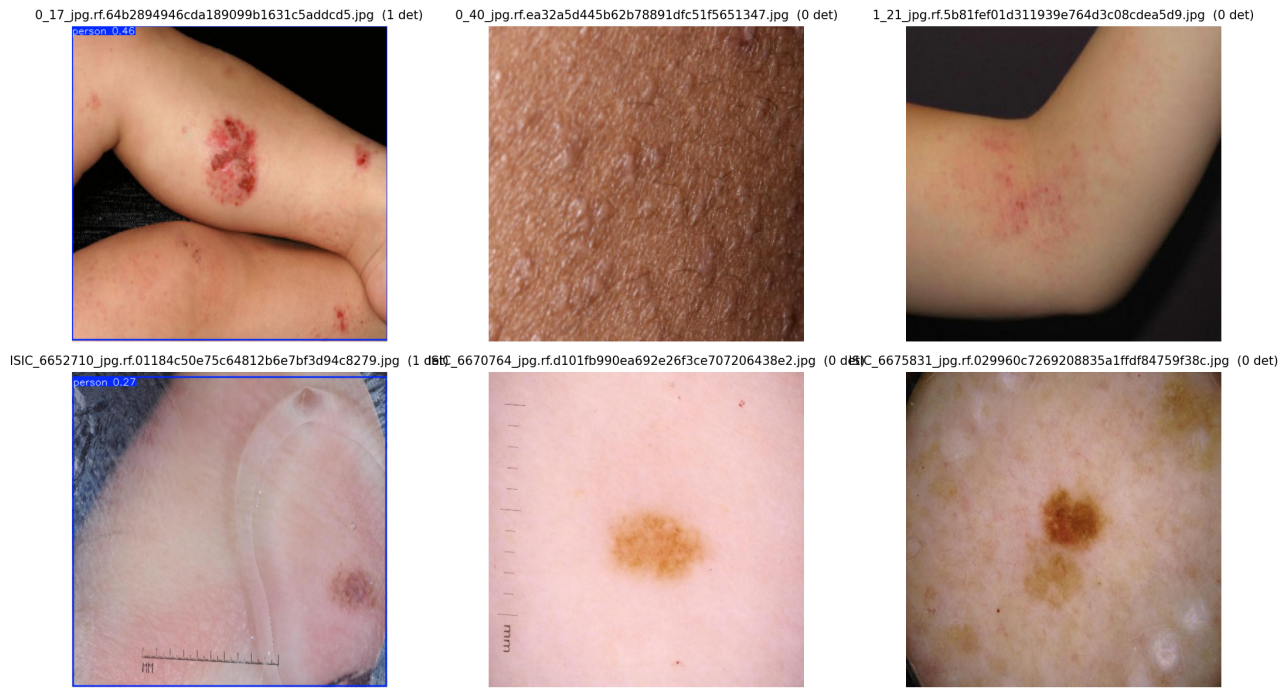


Figure 1: Baseline predictions on 6 example test images. Most images show zero detections; the COCO-pretrained model cannot recognise skin lesions.

Fine-Tuning Setup

Parameter	Value
Base weights	yolo11n.pt (COCO pre-trained)
Epochs	50 (early stop patience = 15)
Batch size	16
Input resolution	640 × 640 px
Frozen layers	None (freeze=0 , full fine-tune)
Device	Apple Silicon (MPS)
Seed	42
Training time	11 minutes

Table 2: Fine-tuning configuration.

Freeze decision: All layers were fine-tuned (**freeze=0**) because the target domain (dermoscopic skin lesions) is visually far from COCO. Freezing the backbone would retain COCO features that do not transfer to subtle lesion boundaries. Full fine-tuning allows the entire network to adapt to the new domain.

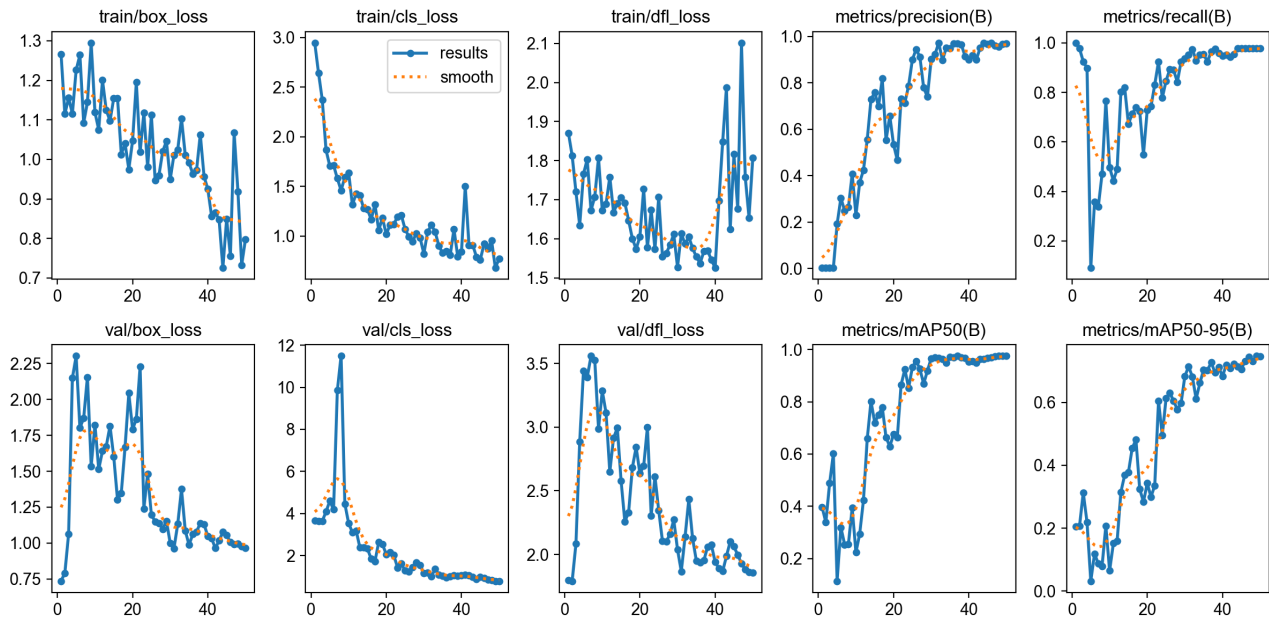


Figure 2: Training curves over 50 epochs. Loss consistently decreases while validation mAP50 converges to ≈ 0.975 on the validation split.

Evaluation and Comparison

Both models were evaluated on the 21-image **test split** using `model.val(split="test")`.

Quantitative Results

Metric	Baseline	Fine-tuned	Δ
Precision (overall)	0.766	0.901	+0.135
Recall (overall)	0.229	0.949	+0.720
mAP50 (overall)	0.260	0.951	+0.691
mAP50-95 (overall)	0.172	0.720	+0.548
mAP50 — Eczema	0.520	0.995	+0.475
mAP50 — melanoma-disease	0.000	0.906	+0.906
mAP50-95 — Eczema	0.344	0.815	+0.471
mAP50-95 — melanoma-disease	0.000	0.625	+0.625

Table 3: Test-split metrics: baseline vs. fine-tuned YOLO11n.

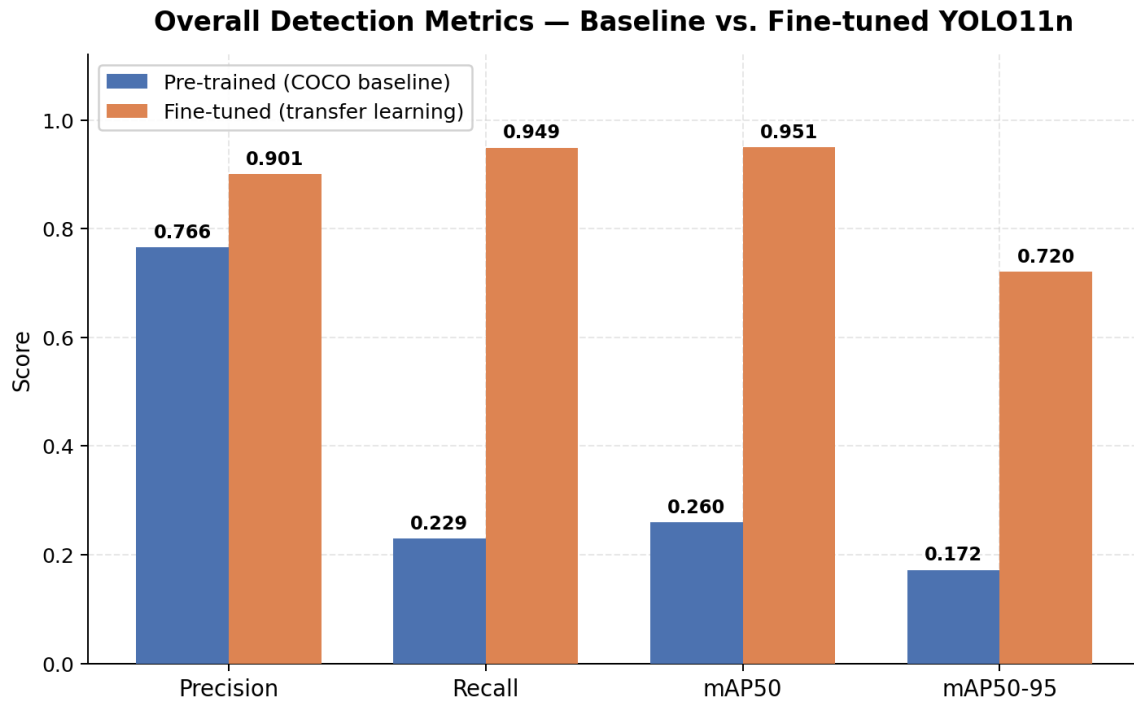
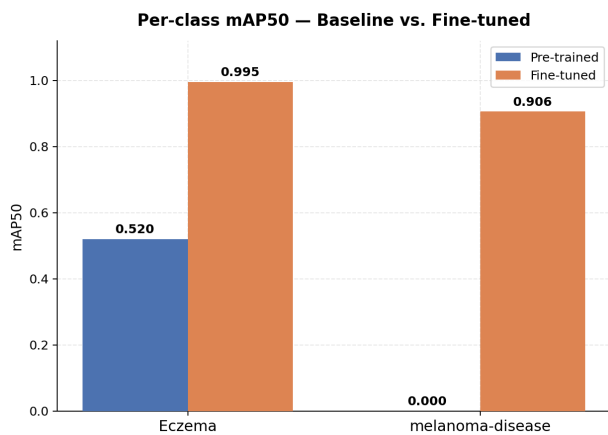
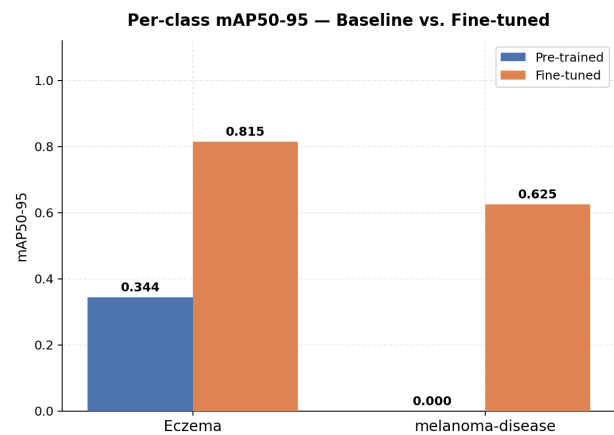


Figure 3: Overall Precision, Recall, mAP50, and mAP50-95 — baseline (left bar) vs. fine-tuned (right bar) for each metric.



(a) Per-class mAP50.



(b) Per-class mAP50-95.

Figure 4: Per-class mAP comparison. `melanoma-disease` improves from 0.000 to 0.906 (mAP50) after fine-tuning.

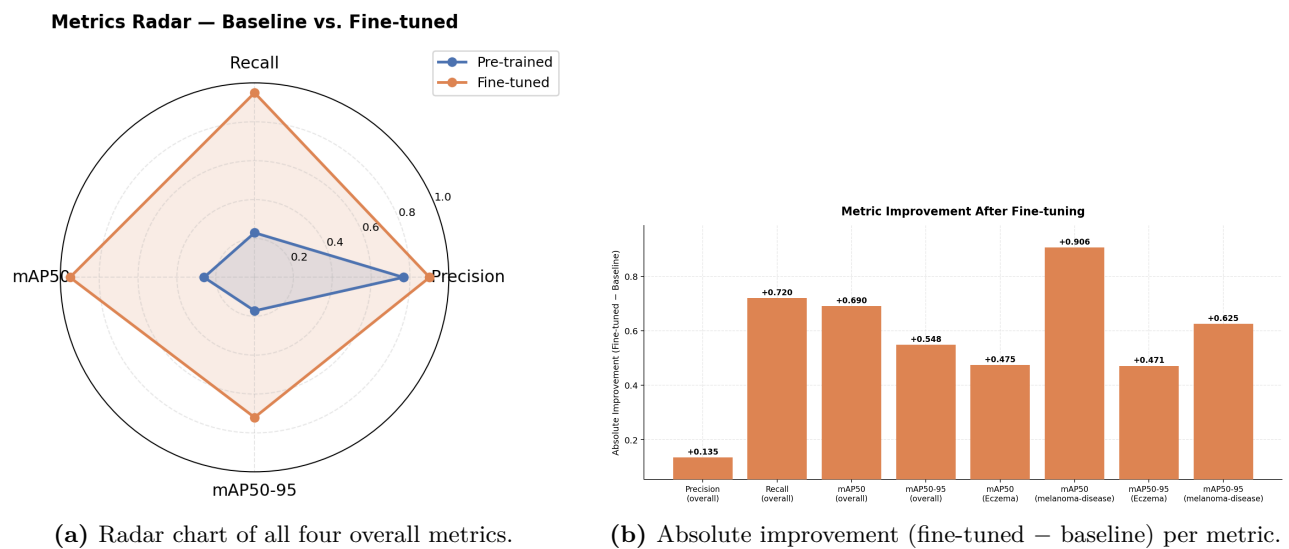


Figure 5: The fine-tuned model dominates the pre-trained baseline across every reported metric.

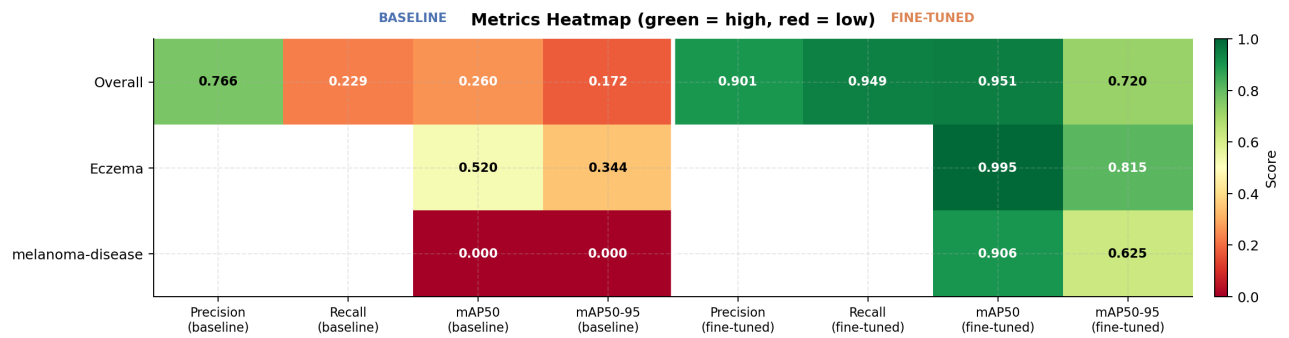


Figure 6: Metrics heatmap (green = high score, red = low score). The fine-tuned columns are uniformly green; the baseline melanoma-disease cells are red (0.000).

Qualitative Comparison

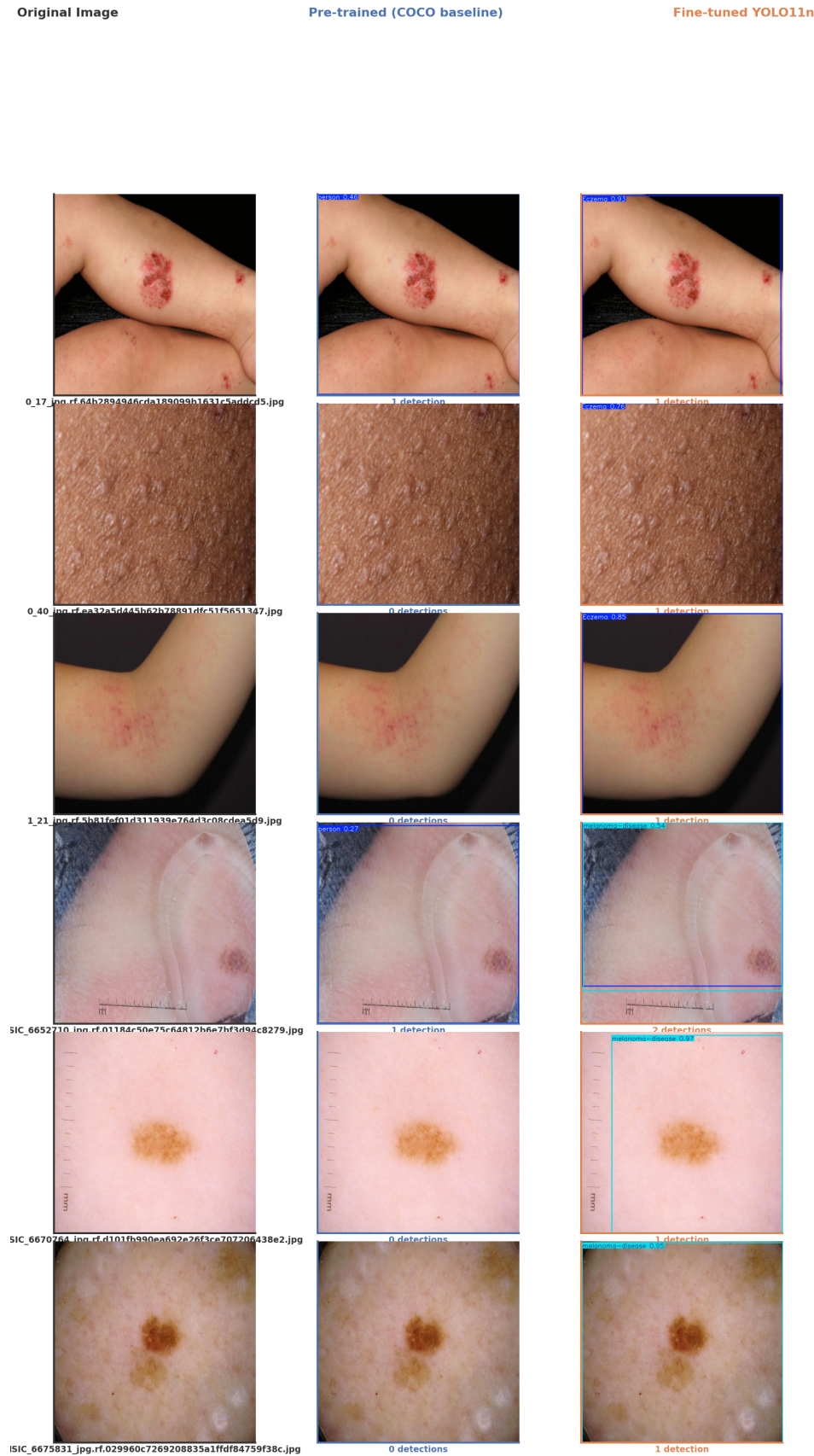


Figure 7: Detection comparison on 6 test images. **Left:** original image. **Centre:** pre-trained (COCO) baseline — mostly empty. **Right:** fine-tuned YOLO11n — correct lesion detections with class labels and confidence scores.

Discussion of improvement and remaining errors:

- **melanoma-disease** goes from $\text{mAP50} = 0.000$ (completely undetectable by the COCO model) to **0.906** — demonstrating the necessity of transfer learning for domain-specific detection tasks.
- **Eczema** improves from 0.520 to **0.995**. The partial baseline score existed because the COCO model occasionally detected similar skin textures under generic classes (e.g. *orange*, *frisbee*).
- **Recall** jumps from 0.229 $\rightarrow$ 0.949: the fine-tuned model finds nearly all lesions, whereas the baseline missed the vast majority.
- **Remaining errors:** occasional confusion between the two lesion types on images where Eczema and melanoma-disease share similar dermoscopic colouring and texture — a known challenge in skin lesion analysis. A handful of false negatives persist on small lesions near image borders.

Gen-AI Reflection

This project used **Claude** (claude-sonnet-4-6, Anthropic) throughout all stages:

- **Planning:** Claude designed the full five-step pipeline and selected the correct Roboflow dataset (discovering that the initially chosen dataset had only one class, then switching to **skin-k1hnx** with two classes).
- **Code generation:** All Python scripts and the Jupyter notebook were scaffolded by Claude and executed end-to-end. Scripts: `download_dataset`, `infer_baseline`, `train_finetune`, `evaluate`, `utils`, `plot_results`.
- **Report:** This $\text{L}^{\text{A}}\text{T}_{\text{E}}\text{X}$ report was drafted by Claude using actual training output, metrics, and generated figures.
- **Verification:** All generated code was executed and outputs (metrics, visualisations) were reviewed against expected behaviour. One bug was corrected during the run (absolute path rewriting in `data.yaml`).

Tools used: Claude (claude.ai/code), Roboflow SDK, Ultralytics YOLO Python API, Tectonic (LaTeX engine), Matplotlib.